

2-Topoi of internal categories

4 Dec 2011

§1. Categories in categories

A category C in a category $\mathcal{C}$ is a diagram

$$C : C_3 \begin{array}{c} \xrightarrow{d_0} \\ \xrightarrow{d_1} \\ \xrightarrow{d_2} \\ \xrightarrow{d_3} \end{array} C_2 \begin{array}{c} \xleftarrow{i_0} \\ \xleftarrow{d_1} \\ \xleftarrow{i_1} \\ \xrightarrow{d_2} \end{array} C_1 \begin{array}{c} \xrightarrow{d_0} \\ \xleftarrow{i} \\ \xrightarrow{d_1} \end{array} C_0$$

in $\mathcal{C}$ such that

$$C1. \quad d_p d_q = d_{q-1} d_p \quad \text{for } p < q$$

$$C2. \quad d_0 i = d_1 i = 1_{C_0}$$

$$d_0 i_0 = d_1 i_0 = d_1 i_1 = d_2 i_1 = 1_{C_1}$$

$$C3. \quad i_0 i = i, i$$

$$d_2 i_0 = i d_1$$

$$d_0 i_1 = i d_0$$

C4. the following squares are pullbacks.

$$\begin{array}{ccc} C_2 & \xrightarrow{d_0} & C_1 \\ d_2 \downarrow & & \downarrow d_1 \\ C_1 & \xrightarrow{d_0} & C_0 \end{array} \quad \begin{array}{ccc} C_3 & \xrightarrow{d_0} & C_2 \\ d_3 \downarrow & & \downarrow d_2 \\ C_2 & \xrightarrow{d_0} & C_1 \end{array}$$

The definition goes back to Ehresmann [].

②

For any object U of $\mathcal{C}$ and any category C in $\mathcal{C}$, we obtain a category $\mathcal{C}(U, C)$ as follows. An object is a morphism $u: U \rightarrow C_0$ in $\mathcal{C}$. A morphism $\gamma: u \rightarrow v$ is a morphism $\gamma: U \rightarrow C_1$ in $\mathcal{C}$ such that $d_0 \gamma = u$ and $d_1 \gamma = v$. The identity $1_u: u \rightarrow u$ is $i_u: U \rightarrow C_1$. The composite of $\gamma: u \rightarrow v$ and $\delta: v \rightarrow w$ is $d_1(\gamma, \delta): U \rightarrow C_1$, where $(\gamma, \delta): U \rightarrow C_2$ is defined by $d_0(\gamma, \delta) = \gamma$ and $d_2(\gamma, \delta) = \delta$.

For each morphism $h: V \rightarrow U$ in $\mathcal{C}$, we have a functor

$$\mathcal{C}(h, C): \mathcal{C}(U, C) \rightarrow \mathcal{C}(V, C)$$

taking u to uh and γ to γh . Thus we obtain a functor

$$\mathcal{C}(-, C): \mathcal{C}^{\text{op}} \rightarrow \text{Cat}.$$

A functor $f: C \rightarrow D$ between categories C and D in $\mathcal{C}$ is a morphism of diagrams

$$F1. \quad \begin{array}{ccccc} C_2 & \xrightleftharpoons[d_2]{d_0} & C_1 & \xrightleftharpoons[d_1]{d_0} & C_0 \\ f_2 \downarrow & & \downarrow f_1 & & \downarrow f_0 \\ D_2 & \xrightleftharpoons[d_2]{d_0} & D_1 & \xrightleftharpoons[d_1]{d_0} & D_0 \end{array}$$

Write $\text{Cat } \mathcal{C}$ for the category of categories

(3)

in $\mathcal{C}$ and functors between them.

Each functor $f : C \rightarrow D$ defines a functor

$$\mathcal{C}(U, f) : \mathcal{C}(U, C) \rightarrow \mathcal{C}(U, D)$$

taking u to $f_0 u$ and γ to $f_1 \gamma$. Thus we obtain a functor

$$\mathcal{C}(-, -) : \mathcal{C}^{\text{op}} \times \text{Cat } \mathcal{C} \rightarrow \text{Cat}$$

which can be recast as a functor

$$\text{Yon}: \text{Cat } \mathcal{C} \longrightarrow [\mathcal{C}^{\text{op}}, \text{Cat}] \quad (1.1)$$

extending the usual Yoneda embedding $\mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \text{Set}]$.

A natural transformation $\theta : f \Rightarrow g : C \rightarrow D$ between functors f and g is a morphism

$$\theta : C_0 \rightarrow D_1$$

in $\mathcal{C}$ such that

$$N1. \quad d_0 \theta = f \text{ and } d_1 \theta = g$$

N2. the following diagram commutes.

$$\begin{array}{ccccc} C_1 & \xrightarrow{(f_1, \theta d_1)} & D_2 & \xrightarrow{d_1} & D_1 \\ & \xrightarrow{(\theta d_0, g_1)} & & & \end{array}$$

(4)

Given a diagram

$$B \xrightarrow{r} C \begin{array}{c} \xrightarrow{f} \\ \Downarrow \theta \\ \xrightarrow{g} \end{array} D \xrightarrow{s} E \quad (1.2)$$

of functors r, f, g, s and a natural transformation θ , we obtain a natural transformation $s\theta r : sfr \Rightarrow sgr$ given by

$$B_0 \xrightarrow{r_0} C_0 \xrightarrow{\theta} D_1 \xrightarrow{s_1} E_1.$$

Given a diagram

$$C \begin{array}{c} \xrightarrow{f} \\ \Downarrow \theta \\ \xrightarrow{g} \end{array} D \begin{array}{c} \xrightarrow{h} \\ \Downarrow \varphi \end{array} \quad (1.3)$$

of functors f, g, h and natural transformations θ, φ , we obtain a natural transformation $\varphi \circ \theta : f \Rightarrow h$ given by

$$C_0 \xrightarrow{(\theta, \varphi)} D_2 \xrightarrow{d_1} D_1.$$

In this way, $\text{Cat } \mathcal{C}$ becomes a 2-category.

Each object U of $\mathcal{C}$ and natural transformation $\theta : f \Rightarrow g : C \rightarrow D$ define a natural transformation

$$\begin{array}{ccc} & \mathcal{C}(U, f) & \\ \mathcal{C}(U, C) & \xrightarrow{\quad \downarrow \mathcal{C}(U, \theta) \quad} & \mathcal{C}(U, D) \\ & \mathcal{C}(U, g) & \end{array}$$

(5)

whose component at $u \in \mathcal{C}(U, C)$ is $\theta u : fu \rightarrow gu$ in $\mathcal{C}(U, D)$. In this way, we see that the extended Yoneda embedding (1.1) is a 2-functor.

For each category C in $\mathcal{C}$, there is a category SC defined as follows. The objects are pairs (U, u) where U is an object of $\mathcal{C}$ and u is an object of $\mathcal{C}(U, C)$. A morphism $(h, \theta) : (U, u) \rightarrow (V, v)$ consists of a morphism $h : U \rightarrow V$ in $\mathcal{C}$ and a morphism $\theta : u \rightarrow v \cdot h$ in $\mathcal{C}(U, C)$. Regarding each object U of $\mathcal{C}$ as a "discrete category" in $\mathcal{C}$ (that is, all $U_n = U$ and $d_p = 1_U$), we see that morphisms of SC are diagrams

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ u \searrow & \xRightarrow{\theta} & \swarrow v \\ & C & \end{array} \quad (1.4)$$

in $\text{Cat } \mathcal{C}$; composition in SC is by pasting these triangles. There is a functor

$$P : SC \longrightarrow \mathcal{C} \quad (1.5)$$

defined by $P(u, u) = u$ and $P(h, \theta) = h$. ⑥

§2. Fibrations and the generalized Yoneda Lemma

Consider a functor $P: \mathcal{X} \rightarrow \mathcal{C}$. A morphism $x: Y \rightarrow X$ in $\mathcal{X}$ is cartesian (with respect to P) when the following square is a pullback for all objects K of $\mathcal{X}$.

$$\begin{array}{ccc} \mathcal{X}(K, Y) & \xrightarrow{\mathcal{X}(K, x)} & \mathcal{X}(K, X) \\ P \downarrow & & \downarrow P \\ \mathcal{C}(PK, PY) & \xrightarrow{\quad} & \mathcal{C}(PK, PX) \\ & \mathcal{C}(PK, Px) & \end{array}$$

Lemma 2.1 Consider a commutative square

$$\begin{array}{ccc} Y' & \xrightarrow{x'} & X' \\ k \downarrow & & \downarrow h \\ Y & \xrightarrow{x} & X \end{array}$$

in $\mathcal{X}$, which is taken to a pullback in $\mathcal{C}$ by the functor $P: \mathcal{X} \rightarrow \mathcal{C}$, and where x is cartesian. The square is a pullback if and only if x' is cartesian.

(7)

Proof Contemplate the two diagrams below.

$$\begin{array}{ccc}
 \mathcal{X}(K, Y') \xrightarrow{\mathcal{X}(1, x')} \mathcal{X}(K, X') & & \mathcal{X}(K, Y') \xrightarrow{\mathcal{X}(1, x')} \mathcal{X}(K, X') \\
 \mathcal{X}(1, k) \downarrow & & \downarrow \mathcal{X}(1, h) \quad P \downarrow \\
 \mathcal{X}(K, Y) \xrightarrow{\mathcal{X}(1, x)} \mathcal{X}(K, X) & & \mathcal{C}(PK, PY') \xrightarrow{\mathcal{C}(1, Px')} \mathcal{C}(PK, PX') \\
 P \downarrow & & \downarrow P \quad \mathcal{C}(1, Ph) \downarrow \\
 \mathcal{C}(PK, PY) \xrightarrow{\mathcal{C}(1, Px)} \mathcal{C}(PK, PX) & & \mathcal{C}(PK, PY) \xrightarrow{\mathcal{C}(1, Px)} \mathcal{C}(PK, PX)
 \end{array}$$

The two squares obtained by composing vertically are equal. By hypothesis, the bottom square in each diagram is a pullback. It follows that the top square in one diagram is a pullback if and only if the top square in the other is. The result follows. $\square$

For any functor $P: \mathcal{X} \rightarrow \mathcal{C}$ and any category C in $\mathcal{C}$, we define a category $\mathcal{X}^C$. The objects are categories X in $\mathcal{X}$ such that $PX = C$ and $d_0: X_1 \rightarrow X_0$ is cartesian. A morphism $r: X \rightarrow Y$ is a functor in $\mathcal{X}$ such that $Pr: PX \rightarrow PY$ is the identity functor of C .

Remark 2.2 By Lemma 2.1, the pullback requirement C4 on a category X in $\mathcal{E}$ belonging to $\mathcal{E}^C$ can be replaced by the requirement that

$d_0 : X_2 \rightarrow X_1$ and $d_0 : X_3 \rightarrow X_2$ should be cartesian.

For functors $P : \mathcal{X} \rightarrow \mathcal{C}$ and $Q : \mathcal{Y} \rightarrow \mathcal{C}$, we write

$$\text{Cat}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$$

for the category of functors $F : \mathcal{X} \rightarrow \mathcal{Y}$ with $QF = P$. The morphisms are natural transformations taken to identities by Q . We write

$$\text{Cart}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$$

for the full subcategory of $\text{Cat}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$ consisting of those functors F which preserve cartesian morphisms. Such an F defines a functor

$$F^C : \mathcal{X}^C \rightarrow \mathcal{Y}^C$$

given by $F^C X = FX$; this uses Remark 2.2.

⑨

We obtain a functor

$$()^C : \text{Cart}_C(\mathcal{X}, \mathcal{Y}) \longrightarrow [\mathcal{X}^C, \mathcal{Y}^C] \quad (2.1)$$

for each category C in $\mathcal{C}$.

A functor $P : \mathcal{X} \rightarrow \mathcal{C}$ is called a fibration [Groth] when, for all $h : V \rightarrow U$ in $\mathcal{C}$ and X in $\mathcal{X}$ with $PX = U$, there exists a cartesian morphism $x : Y \rightarrow X$ with $Px = h$. Such an $x : Y \rightarrow X$ is called an inverse image of X along h . A choice of inverse image for all such X and h is called a cleavage for P ; we write

$$h^X : h^*X \rightarrow X$$

for the chosen cartesian morphisms. Given a cleavage, $\mathcal{X}^C$ will be restricted to those E with $d_0 : E_{n+1} \rightarrow E_n$ a chosen cartesian morphism.
~~chosen cartesian morphism~~ let us now return to the functor $P : \mathcal{SC} \rightarrow \mathcal{C}$

of (1.5). A morphism $(h, \theta) : (U, u) \rightarrow (V, v)$ is cartesian if and only if $\theta : u \rightarrow vh$ is invertible in the category $\mathcal{C}(U, C)$. In fact, $P : \mathcal{SC} \rightarrow \mathcal{C}$ is a fibration and there is a cleavage for which the chosen cartesian (h, θ) have θ an identity.

(10)

The following diagram defines a category $\hat{C}$ in SC :

$$\begin{array}{ccccc}
 & \xrightarrow{(d_0, id_{d_0})} & & \xrightarrow{(d_0, id_0)} & \\
 (C_2, d_0 d_0) & \xrightarrow{(d_1, id_{d_0 d_0})} & (C_1, d_0) & \xleftarrow{(i, i)} & (C_0, 1_{C_0}) \\
 & \xrightarrow{(d_2, d_0)} & & \xrightarrow{(d_1, 1_{C_1})} &
 \end{array}$$

For each object (U, u) of SC , we can identify the category

$$SC((U, u), \hat{C}) = u/\mathcal{C}(U, C) \quad (2.2)$$

as the category of objects of $\mathcal{C}(U, C)$ under the object u .

Note moreover that $\hat{C}$ is actually an object of $(SC)^C$. The generalised Yoneda Lemma of [ST Pub.13; Theorem 5.15] expresses the "generic" property of $\hat{C} \in (SC)^C$.

Theorem 2.2 For any fibration $P: \mathcal{X} \rightarrow \mathcal{C}_0$ and category C in $\mathcal{C}$, the composite functor

$$Cart_{\mathcal{C}}(SC, \mathcal{X}) \xrightarrow{(\cdot)^C} [(SC)^C, \mathcal{X}^C] \xrightarrow{eval_{\hat{C}}} \mathcal{X}^C$$

is an equivalence of categories.

Proof Take $X \in \mathcal{X}^C$. We must define a cartesian-(11)
morphism-preserving functor

$$\bar{X} : \mathcal{SC} \longrightarrow \mathcal{X}$$

over $\mathcal{C}$. For $(U, u) \in \mathcal{SC}$, put

$$\bar{X}(U, u) = u^* X_0.$$

For $(h, \theta) : (U, u) \longrightarrow (V, v)$ in $\mathcal{SC}$, notice that the composite of cartesian morphisms

$$\theta^* X_1 \xrightarrow{\theta^{X_1}} X_1 \xrightarrow{d_0} X_0$$

is cartesian over $u; U \longrightarrow C_0$ (since $d_0 \theta = u$), so we have an isomorphism $u^* X_0 \cong \theta^* X_1$ over U and commuting with u^{X_0} and $d_0 \theta^{X_1}$. Define $\bar{X}(h, \theta) : \bar{X}(U, u) \longrightarrow \bar{X}(V, v)$ such that

$$\begin{array}{ccc} v^* X_0 & \xrightarrow{v^{X_0}} & X_0 \\ \bar{X}(h, \theta) \uparrow & & \uparrow d_1 \\ u^* X_0 \cong \theta^* X_1 & \xrightarrow{\theta^{X_1}} & X_1 \end{array}$$

commutes and $P \bar{X}(h, \theta) = h$. Please see [13; Theorem 5.15] for further details. $\square$

A split fibration is a fibration equipped with a cleavage satisfying

$$\begin{aligned} 1_u^* X &= X, & 1_u^X &= 1_X \\ (uv)^* X &= v^* u^* X, & (uv)^X &= u^X v^{u^* X}. \end{aligned}$$

(12)

Write $\text{Spl}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$ for the full subcategory of $\text{Cart}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y})$ consisting of those $F: \mathcal{X} \rightarrow \mathcal{Y}$ which preserve cleavages given in $\mathcal{X}$ and $\mathcal{Y}$. When $\mathcal{X}$ is equipped with a cleavage, we understand that $\mathcal{X}^{\mathcal{C}}$ consists of those X for which the $d_0: X_{n+1} \rightarrow X_n$ are in the cleavage.

Notice that $P: \text{SC} \rightarrow \mathcal{C}$ is a split fibration.

Theorem 5.15 of [StRut, 13] also included the following (recalling our convention about $\mathcal{X}^{\mathcal{C}}$ in the cloven case):

Proposition 2.3 For any split fibration $P: \mathcal{X} \rightarrow \mathcal{C}$ and category $\mathcal{C}$ in $\mathcal{C}$, the equivalence of Theorem 2.2 restricts to an isomorphism of categories

$$\text{Spl}_{\mathcal{C}}(\text{SC}, \mathcal{X}) \cong \mathcal{X}^{\mathcal{C}}. \quad \square$$

Corollary 2.4 The 2-functor

$$\text{Yon} : \text{Cat } \mathcal{C} \longrightarrow [\mathcal{C}^{\text{op}}, \text{Cat}]$$

is fully faithful. That is, the inclusion 2-functor

$$\mathcal{C} \longrightarrow \text{Cat } \mathcal{C}$$

is dense.

(13)

Proof For categories C and D in $\mathcal{C}$, we have a commutative square

$$\begin{array}{ccc} (SD)^C & \xrightarrow{\cong} & Spl_{\mathcal{C}}(SC, SD) \\ \cong \downarrow & & \downarrow \cong \\ Cat_{\mathcal{C}}(C, D) & \xrightarrow{\gamma_{C,D}} & [\mathcal{C}^{\text{op}}, Cat](\mathcal{C}(C, -), \mathcal{C}(D, -)) \end{array}$$

where the vertical isomorphisms are fairly straightforward and the top isomorphism comes from Proposition 2.3. $\square$

Remark 2.5 Perhaps the reader noticed that $d_1: \hat{C}_2 \rightarrow \hat{C}_1$ in the generic example is actually cartesian, in fact, part of the cleavage. It follows from Theorem 2.2 that, for any $X \in \mathcal{C}$, the morphism $d_1: X_2 \rightarrow X_1$ is cartesian. This should not surprise since $d_0 d_1 = d_0 d_0$ and we have diagrams

$$\begin{array}{ccccccc} \mathcal{X}(K, X_2) & \xrightarrow{\mathcal{X}(1, d_1)} & \mathcal{X}(K, X_1) & \xrightarrow{\mathcal{X}(1, d_0)} & \mathcal{X}(K, X_0) & & \\ p \downarrow & & \downarrow p & pb & \downarrow p & & \\ \mathcal{C}(PK, C_2) & \xrightarrow{\mathcal{C}(1, d_1)} & \mathcal{C}(PK, C_1) & \xrightarrow{\mathcal{C}(1, d_0)} & \mathcal{C}(PK, C_0) & & \\ \mathcal{X}(K, X_2) & \xrightarrow{\mathcal{X}(1, d_0)} & \mathcal{X}(K, X_1) & \xrightarrow{\mathcal{X}(1, d_0)} & \mathcal{X}(K, X_0) & & \\ p \downarrow & pb & p \downarrow & pb & \downarrow p & & \\ \mathcal{C}(PK, C_2) & \xrightarrow{\mathcal{C}(1, d_0)} & \mathcal{C}(PK, C_1) & \xrightarrow{\mathcal{C}(1, d_0)} & \mathcal{C}(PK, C_0) & & \end{array}$$

(146)

from which it follows that the top left square is a pullback, yielding that $d_1 : X_2 \rightarrow X_1$ is cartesian.

We conclude this section with some facts all due to Grothendieck [Groth.].

A cleavage for a fibration $P : \mathcal{X} \rightarrow \mathcal{C}$ allows the definition of a pseudofunctor $\mathcal{X}^- : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$. The value at $U \in \mathcal{C}$ is the fibre $\mathcal{X}^U$ of P over U : it is the subcategory of $\mathcal{X}$ consisting of the objects $X \in \mathcal{X}$ with $PX = U$ and morphisms $x : Y \rightarrow X$ with $Px = 1_U$. (This agrees with the previous notation $\mathcal{X}^C$ when $C = U$ is discrete.) The value of the pseudofunctor at $h : V \rightarrow U$ in $\mathcal{C}$ is the functor

$$h^* : \mathcal{X}^U \rightarrow \mathcal{X}^V$$

taking X to h^*X and using the universal property of cartesian morphisms to define h^* on morphisms. If $P : \mathcal{X} \rightarrow \mathcal{C}$ is a split fibration, $\mathcal{X}^- : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$ is a functor. If $\text{Spl}_{\mathcal{C}}$ is the 2-category of split fibrations over $\mathcal{C}$ and cleavage preserving functors over $\mathcal{C}$ then we obtain an equivalence of 2-categories

$$\text{Spl}_{\mathcal{C}} \simeq [\mathcal{C}^{\text{op}}, \text{Cat}] \quad (2.3)$$

taking $P : \mathcal{X} \rightarrow \mathcal{C}$ to $\mathcal{X}^-$.

§3. Internal full subcategories

Suppose $\mathcal{C}$ is a category admitting pullbacks. Let $\mathcal{C}^2 = [\mathcal{I}, \mathcal{C}]$ be the category of morphisms in $\mathcal{C}$. The functor

$$\text{cod} : \mathcal{C}^2 \longrightarrow \mathcal{C},$$

taking each morphism to its codomain, is a fibration. The cartesian morphisms in $\mathcal{C}^2$ are the pullback squares in $\mathcal{C}$.

For any category C in $\mathcal{C}$, the category $(\mathcal{C}^2)^C$ has objects those functors $p : E \rightarrow C$ in $\mathcal{C}$ such that the square

$$\begin{array}{ccc} E_1 & \xrightarrow{d_0} & E_0 \\ p_1 \downarrow & & \downarrow p_0 \\ C_1 & \xrightarrow{d_0} & C_0 \end{array} \quad (3.1)$$

is a pullback. Such functors p are called discrete offibrations over C . We think of

them as internalized functors $C \rightarrow \mathcal{C}$.

By Theorem 2.2, each discrete offibration $p : E \rightarrow C$ corresponds to a cartesian-morphism-preserving functor

$$\bar{p} : SC \longrightarrow \mathcal{C}^2$$

taking (U, u) to $p_u: E_u \rightarrow U$ where the square (16)

$$\begin{array}{ccc} E_u & \xrightarrow{i_u} & E \\ p_u \downarrow & & \downarrow p \\ U & \xrightarrow{u} & C \end{array} \quad (3.2)$$

is a pullback. For a morphism $(h, \theta): (U, u) \rightarrow (V, v)$, we obtain the morphism $\bar{p}(h, \theta):$

$$\begin{array}{ccc} E_u & \xrightarrow{\bar{\theta}} & E_v \\ p_u \downarrow & & \downarrow p_v \\ U & \xrightarrow{h} & V \end{array} \quad (3.3)$$

in $\mathcal{E}^2$ using the pullback (3.2) for v and the diagram

$$\begin{array}{ccccc} E_u & \xrightarrow{d_1(\theta p_u, p_v i_u)} & E & & \\ p_u \downarrow & & \downarrow p & & \\ U & \xrightarrow{h} & V & \xrightarrow{v} & C \end{array}$$

for which the pullback (3.1) is used to define

$(\theta p_u, p_v i_u): E_u \rightarrow E_1$ satisfying

$$d_0(\theta p_u, p_v i_u) = i_u \quad \text{and}$$

$$p_1(\theta p_u, p_v i_u) = \theta p_u.$$

The split fibration $P: SC \rightarrow \mathcal{C}$ corresponds to the functor

$$\mathcal{C}(-, C) : \mathcal{C}^{\text{op}} \longrightarrow \text{Cat}.$$

A choice of pullbacks in $\mathcal{C}$ means that the fibration $\text{cod} : \mathcal{C}^2 \rightarrow \mathcal{C}$ leads to a pseudofunctor

$$\mathcal{C}/- : \mathcal{C}^{\text{op}} \longrightarrow \text{Cat}$$

taking each object U to the slice category $\mathcal{C}/U$ and defined on morphisms by pullback along them. It follows that $\bar{P}$ induces a pseudonatural transformation with component at $U \in \mathcal{C}$ denoted by

$$\bar{P}^U : \mathcal{C}(U, C) \longrightarrow \mathcal{C}/U. \quad (3.4)$$

Proposition 3.1 A functor

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{F} & \mathcal{Y} \\ & P \searrow & \swarrow Q \\ & \mathcal{C} & \end{array} \quad (3.5)$$

over $\mathcal{C}$ induces a family of functors

$$F^U : \mathcal{X}^U \longrightarrow \mathcal{Y}^U, \quad U \in \mathcal{C}, \quad (3.6)$$

on the fibres. If F is fully faithful then so are all the functors F^U , $U \in \mathcal{C}$. The converse holds if P is a fibration and F preserves cartesian morphisms.

Proof If $y: FX \rightarrow FX'$ in $\mathcal{Y}$ is in $\mathcal{Y}^u$ then any $x: X \rightarrow X'$ in $\mathcal{X}$ with $Fx = y$ is in $\mathcal{X}^u$. This proves the second sentence of the Proposition. For the converse, take any $h: U \rightarrow U'$ in $\mathcal{C}$ and $x: Z \rightarrow X'$ cartesian with $Px = h$. We have a diagram of pullbacks:

$$\begin{array}{ccccc} \mathcal{X}^u(X, Z) & \longrightarrow & \mathcal{X}(X, Z) & \xrightarrow{\mathcal{X}(1, x)} & \mathcal{X}(X, X') \\ \downarrow F^u & \lrcorner & \downarrow F & \lrcorner & \downarrow F \\ \mathcal{Y}^u(FX, FZ) & \longrightarrow & \mathcal{Y}(FX, FZ) & \xrightarrow{\mathcal{Y}(1, Fx)} & \mathcal{Y}(FX, FX') \\ \downarrow & \lrcorner & \downarrow Q & \lrcorner & \downarrow Q \\ 1 & \xrightarrow{1_u} & \mathcal{C}(U, U) & \xrightarrow{\mathcal{C}(1, h)} & \mathcal{C}(U, U') \end{array}$$

By assumption, the top left vertical function is invertible. So F is fully faithful on those morphisms over h ; but all morphisms are over some h . So F is fully faithful. $\square$

An internal full subcategory of the category $\mathcal{C}$ is a discrete opfibration $\bar{p}: E \rightarrow \mathcal{C}$ for which each $\bar{p}^u$ (3.4) is fully faithful. By Proposition 3.1, this is the same as the requirement that the functor $\bar{p}: SC \rightarrow \mathcal{C}^2$ be fully faithful.

For any discrete offibration $p: E \rightarrow C$, we have a factorization

$$\begin{array}{ccc} \mathcal{C}/C_0 & \xrightarrow{\bar{p}_0} & \mathcal{C}^{\mathcal{D}} \\ & \searrow & \nearrow \bar{p} \\ & SC & \end{array} \quad (3.5)$$

of $\bar{p}_0$ over $\mathcal{C}$ in which the first functor is bijective on objects. All three functors preserve cartesian morphisms; indeed, $\mathcal{C}/C_0 \rightarrow SC$ preserves the cleavages.

For any cartesian-morphism-preserving functor $F:$

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{F} & \mathcal{Y} \\ & \searrow P & \swarrow Q \\ & \mathcal{C} & \end{array} \quad (3.6)$$

over $\mathcal{C}$, we can take the "full image"

$$\begin{array}{ccccc} \mathcal{X} & \xrightarrow{H} & \mathcal{Z} & \xrightarrow{G} & \mathcal{Y} \\ & \searrow P & \downarrow R & \swarrow Q & \\ & & \mathcal{C} & & \end{array} \quad (3.7)$$

where $\mathcal{Z}$ is the category with the same objects as $\mathcal{X}$ and with homsets $\mathcal{Z}(X', X) = \mathcal{Y}(FX', FX)$. The functor H is the

identity on objects and defined by

$$\mathcal{X}(X', X) \xrightarrow{F} \mathcal{Y}(FX', FX) = \mathcal{Z}(X', X)$$

on morphisms. The functor G is defined to be F on objects and the identity on morphisms.

The functor R is unique making (3.7) commute. The cartesian morphisms in $\mathcal{Z}$ are those which are cartesian in $\mathcal{Y}$. If P is a fibration then so is R ; given $h: V \rightarrow U$ in $\mathcal{C}$ and $X \in \mathcal{Z}$ with $RX = PX = U$, there is a cartesian $x: X' \rightarrow X$ over h for P , and $Fx: X' \rightarrow X$ in $\mathcal{Z}$ is cartesian over h for R . Both H and G preserve cartesian morphisms.

Let us write $P_F: \mathcal{X}[F] \rightarrow \mathcal{C}$ for the functor $R: \mathcal{Z} \rightarrow \mathcal{C}$ obtained using the above full image construction.

The 2-category $\text{Cat}/\mathcal{C}$ is complete. In particular, we are interested in the comma object $F/_\mathcal{C} F'$ of two functors

$$\begin{array}{ccccc} \mathcal{X} & \xrightarrow{F} & \mathcal{Y} & \xleftarrow{F'} & \mathcal{X}' \\ & \searrow P & \downarrow Q & \swarrow P' & \\ & & \mathcal{C} & & \end{array} \quad (3.8)$$

Indeed, $F/_\mathcal{C} F'$ is the full subcategory of the comma category F/F' consisting of those objects (X, y, X') , where $X \in \mathcal{X}$, $X' \in \mathcal{X}'$ and $y \in \mathcal{Y}(FX, F'X')$, for which Qy is an identity morphism in $\mathcal{C}$. We have a universal natural transformation

$$\begin{array}{ccc} F/_\mathcal{C} F' & \xrightarrow{\text{cod}} & \mathcal{X}' \\ \text{dom} \downarrow & \xRightarrow{\lambda} & \downarrow F' \\ \mathcal{X} & \xrightarrow{F} & \mathcal{Y} \end{array}$$

over $\mathcal{C}$ (that is, identified by $Q: \mathcal{Y} \rightarrow \mathcal{C}$). If F and F' preserve cartesian morphisms then so do dom and cod , and the cartesian morphisms of $F/_\mathcal{C} F'$ are precisely those taken to cartesian morphisms by dom and cod .

Proposition 3.2 The 2-functor

$$J : \text{Cat } \mathcal{C} \longrightarrow \text{Cat}/\mathcal{C}$$

preserves comma objects.

Proof The universal property of a comma object

$$\begin{array}{ccc} f/f' & \xrightarrow{d'} & C' \\ d \downarrow & \xRightarrow{\lambda} & \downarrow f' \\ C & \xrightarrow{f} & D \end{array}$$

in $\text{Cat } \mathcal{C}$ immediately yields that the functor

$$S(f/f') \longrightarrow Sf/_e S f',$$

taking the morphism

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ u \searrow & \xRightarrow{\theta} & \swarrow v \\ & f/f' & \end{array}$$

in $S(f/f')$ to the morphism

$$((h, d\theta), (h, d'\theta)) : ((U, du), \lambda u, (U, d'u)) \rightarrow ((V, dv), \lambda v, (V, d'v))$$

in $Sf/_e S f'$, is an isomorphism. $\square$

A morphism $k; M \rightarrow N$ in $\mathcal{C}$ (can be regarded as a discrete opfibration between discrete categories in $\mathcal{C}$ and so) gives rise to a cartesian-morphism-preserving functor

$$\begin{array}{ccc} \mathcal{C}/N & \xrightarrow{\bar{k}} & \mathcal{C}^2 \\ \text{dom} \searrow & & \swarrow \text{cod} \\ & \mathcal{C} & \end{array}$$

(3.9)

over $\mathcal{C}$. We can factor $\bar{k}$ as

$$\begin{array}{ccccc}
 \mathcal{C}/N & \xrightarrow{H} & (\mathcal{C}/N)[\bar{k}] & \xrightarrow{G} & \mathcal{C}^2 \\
 & \searrow \text{dom} & \downarrow P_{\bar{k}} & \swarrow \text{cod} & \\
 & & \mathcal{C} & &
 \end{array} \quad (3.10)$$

where H is the identity on objects and G is fully faithful. The objects of $(\mathcal{C}/N)[\bar{k}]$ are pairs (U, u) for $u: U \rightarrow N$ in $\mathcal{C}$ and the morphisms are commutative squares

$$\begin{array}{ccc}
 M_u & \longrightarrow & M_v \\
 k_u \downarrow & & \downarrow k_v \\
 U & \xrightarrow{h} & V
 \end{array} \quad (3.11)$$

where k_u is the pullback of k along u .

Proposition 3.3 [St 13; Thm 6.2] For a morphism $k: M \rightarrow N$ in a finitely complete category $\mathcal{C}$, the following three conditions are equivalent:

- (i) there exists an internal full subcategory $p: \mathcal{E} \rightarrow \mathcal{C}$ of $\mathcal{C}$ with $p_0 = k$;
- (ii) there exists a category $\mathcal{C}$ in $\mathcal{C}$ with $\mathcal{C}_0 = N$ and an isomorphism

$$|\mathcal{C}| \cong (\mathcal{C}/N)[\bar{k}]$$

of categories over $\mathcal{C}$;

- (iii) there exists a cartesian internal hom for the two objects

$$k \times 1: M \times N \rightarrow N \times N \quad \text{and} \quad 1 \times k: N \times M \rightarrow N \times N$$

of the category $\mathcal{C}/N \times N$.

Proof (i) $\Rightarrow$ (ii) Given (i), compare the factorizations (3.5) and (3.10). Since bijective-on-objects and fully faithful functors form a factorization system on Cat , the two factorizations are isomorphic, yielding (ii).

(ii) $\Rightarrow$ (iii) The product of $(u, v): U \rightarrow N \times N$ and $k \times 1: M \times N \rightarrow N \times N$ is the main diagonal of the pullback

$$\begin{array}{ccc} M_u & \longrightarrow & M \times N \\ k_u \downarrow & & \downarrow k \times 1 \\ U & \xrightarrow{(u, v)} & N \times N \end{array} \quad \text{in } \mathcal{C}/N \times N$$

in $\mathcal{C}$. So

$$(\mathcal{C}/N \times N) \left(\begin{array}{c} U \times_{N \times N} (M \times N) \\ \downarrow \\ N \times N \end{array}, \begin{array}{c} N \times M \\ \downarrow 1 \times k \\ N \times N \end{array} \right) \cong (\mathcal{C}/U) \left(\begin{array}{c} M_u \\ \downarrow k_u \\ U \end{array}, \begin{array}{c} M_v \\ \downarrow k_v \\ U \end{array} \right). \quad (3.12)$$

On the other hand, for any category C with $C_0 = N$, we have

$$(\mathcal{C}/N \times N) \left(\begin{array}{c} U \\ \downarrow (u, v) \\ N \times N \end{array}, \begin{array}{c} C_1 \\ \downarrow (d_0, d_1) \\ N \times N \end{array} \right) \cong \mathcal{C}(U, C)(u, v). \quad (3.13)$$

The isomorphism of (ii) yields an isomorphism between the right hand side of (3.12) and the right hand side of (3.13). This shows that

$$(d_0, d_1): C_1 \rightarrow N \times N$$

is an internal hom as required for (iii).

(ii) $\Rightarrow$ (i) Let $(d_0, d_1): C_1 \rightarrow N \times N$ be an internal 25
 hom as in (iii) and let $C_0 = N$. Despite C not yet
 being a category, only a graph, the left hand
 side of (3.12) and that of (3.13) are isomorphic.
 Therefore, looking at the right hand sides, we
 have a fully faithful graph morphism

$$\mathcal{C}(U, C) \longrightarrow \mathcal{C}/U, \quad u \mapsto k_u.$$

Since the codomain is a category, a category
 structure is induced on $\mathcal{C}(U, C)$. The family
 of functors is pseudonatural in U and so
 yields a fully faithful cartesian-morphism-
 preserving functor $\mathcal{S}C \rightarrow \mathcal{C}^2$ over $\mathcal{C}$.
 By Theorem 2.2, (i) follows. $\square$

§4. Descent matters

(26)

As usual we write Δ for the (topologists') simplicial category. The objects are the non-empty ordered sets

$$[n] = \{0, 1, \dots, n\}$$

and the morphisms are order-preserving functions. We write

$$\partial_p : [n-1] \longrightarrow [n] \quad \text{and} \quad \sigma_q : [n+1] \longrightarrow [n]$$

for the injective order-preserving function which omits p in its image and the surjective order-preserving function which identifies only q and $q+1$. The squares

$$\begin{array}{ccc} [n-1] & \xrightarrow{\partial_0} & [n] \\ \partial_n \downarrow & & \downarrow \partial_{n+1} \\ [n] & \xrightarrow{\partial_0} & [n+1] \end{array}$$

are pushouts in Δ .

We write $d_p : [n] \rightarrow [n-1]$ and $i_q : [n] \rightarrow [n+1]$ for ∂_p and σ_q when thinking of them as morphisms of Δ^{op} .

Let $\mathbb{D}$ denote the category

$$\begin{array}{ccccc}
 & \xrightarrow{d_0} & & \xrightarrow{d_0} & & \xrightarrow{d_0} & \\
 [3] & \xrightarrow{d_1} & [2] & \xleftarrow{i_0} & [1] & \xleftarrow{i_0} & [0] \\
 & \xrightarrow{d_2} & & \xleftarrow{d_1} & & \xleftarrow{d_1} & \\
 & \xrightarrow{d_3} & & \xleftarrow{c_1} & & & \\
 & & & \xrightarrow{d_2} & & &
 \end{array}$$

in Δ^{op} .

Definition 4.1 The descent category for a functor $P : \mathcal{E} \rightarrow \Delta^{\text{op}}$ is

$$\text{Desc } \mathcal{E} = \mathcal{E}^{\mathbb{D}}.$$

Objects E of $\text{Desc } \mathcal{E}$ are called descent data for P .

In Definition 4.1, if P is a fibration, a cleavage determines a pseudofunctor

$$\mathcal{E}^- : \Delta \rightarrow \text{Cat};$$

that is, a pseudocosplicial category.

We write $\mathcal{E}_n$ for the fibre $\mathcal{E}^{[n]}$ of P over $[n]$, yielding a diagram

$$\begin{array}{ccccccc}
 \mathcal{E}_0 & \xrightarrow{\partial_0} & \mathcal{E}_1 & \xrightarrow{\partial_0} & \mathcal{E}_2 & \xrightarrow{\partial_0} & \mathcal{E}_3 \\
 & \xleftarrow{\partial_1} & & \xleftarrow{\partial_1} & & \xleftarrow{\partial_1} & \\
 & & \mathcal{E}_1 & \xleftarrow{\partial_2} & \mathcal{E}_2 & \xleftarrow{\partial_2} & \mathcal{E}_3 \\
 & & & \xleftarrow{\partial_2} & & \xleftarrow{\partial_2} & \\
 & & & & \mathcal{E}_2 & \xleftarrow{\partial_3} & \mathcal{E}_3
 \end{array}$$

in Cat

in which the simplicial identities hold up to coherent isomorphism: for example,

$$\partial_q \partial_p \cong \partial_p \partial_{q-1} \quad \text{for } p < q.$$

For descent data E , define

$$e : \partial_0 E_0 \longrightarrow \partial_1 E_0 \quad (4.1)$$

to be the unique morphism in $\mathcal{C}_1$ such that

$$E_1 = \partial_0 E_0 \begin{array}{c} \xrightarrow{e} \partial_1 E_0 \\ \xrightarrow{d_1} E_0 \end{array} \begin{array}{c} \xrightarrow{d_1^{E_0}} \\ \end{array}$$

commutes. We readily see that the diagram

$$\begin{array}{ccc} \partial_1 \partial_0 E_0 & \xrightarrow{\partial_1 e} & \partial_1 \partial_1 E_0 \\ \parallel & & \cong \\ \partial_0 \partial_0 E_0 & \xrightarrow{\partial_0 e} & \partial_0 \partial_1 E_0 \cong \partial_2 \partial_0 E_0 \end{array} \begin{array}{c} \xrightarrow{\partial_2 e} \\ \end{array} \partial_2 \partial_1 E_0 \quad (4.2)$$

commutes in $\mathcal{C}_2$ and

$$\begin{array}{ccc} \sigma_0 \partial_0 E_0 & \xrightarrow{\sigma_0 e} & \sigma_0 \partial_1 E_0 \\ \cong & & \cong \\ & E_0 & \end{array} \quad (4.3)$$

commutes in $\mathcal{C}_0$. In fact, we can reconstruct E from E_0 and e satisfying (4.2) and (4.3).

While we have defined the descent category (for a category "parametrized" by Δ^{op}) in terms of the $\mathcal{X}^C$ construction, we can go the other way too. Suppose $P: \mathcal{X} \rightarrow \mathcal{C}$ is a functor and C is a category in $\mathcal{C}$. For simplicity we suppose $\mathcal{C}$ has pullbacks chosen so that we can complete the diagram for C to a simplicial object (functor)

$$C: \Delta^{\text{op}} \rightarrow \mathcal{C}.$$

Form the pullback

$$\begin{array}{ccc} C^* \mathcal{X} & \longrightarrow & \mathcal{X} \\ P_C \downarrow & & \downarrow P \\ \Delta^{\text{op}} & \xrightarrow{C} & \mathcal{C} \end{array} \quad (4.4)$$

in Cat .

Proposition 4.2 There is a canonical equivalence of categories

$$\mathcal{X}^C \simeq \text{Desc } C^* \mathcal{X}$$

$$E \longleftrightarrow (\mathbb{D}, E).$$

For cloven P , this is an isomorphism.

(30)

For any functor $P: \mathcal{E} \rightarrow \mathcal{C}$ and category $\mathcal{H}$, the cotensor of $\mathcal{H}$ with P in the 2-category $\text{Cat}/\mathcal{C}$ is defined by the pullback

$$\begin{array}{ccc} [\mathcal{H}, \mathcal{E}]_{\mathcal{C}} & \longrightarrow & [\mathcal{H}, \mathcal{E}] \\ p^{\mathcal{H}} \downarrow & & \downarrow [1, P] \\ \mathcal{C} & \xrightarrow{\text{diag}} & [\mathcal{H}, \mathcal{C}] \end{array}$$

in Cat . For a cloven fibration P , the pseudo-functor corresponding to the cloven fibration $p^{\mathcal{H}}$ is the composite

$$\mathcal{C}^{\text{op}} \xrightarrow{\mathcal{E}^-} \text{Cat} \xrightarrow{[\mathcal{H}, -]} \text{Cat}.$$

Proposition 4.3 For any functor $P: \mathcal{E} \rightarrow \Delta^{\text{op}}$
and category $\mathcal{H}$, there is a canonical isomorphism of categories

$$[\mathcal{H}, \text{Desc } \mathcal{E}] \cong \text{Desc} [\mathcal{H}, \mathcal{E}]_{\Delta^{\text{op}}}.$$

Loosely speaking, this says that the descent construction is preserved by representable 2-functors out of Cat . It therefore makes sense to define the construction in a 2-category $\mathcal{K}$.

For a pseudofunctor $T: \Delta \rightarrow \text{Cat}$, we write $\text{Desc } T$ for the category $\text{Desc } \mathcal{E}$ where $P: \mathcal{E} \rightarrow \Delta^{\text{op}}$ is the cloven fibration obtained from T by the Grothendieck construction.

Definition 4.4 For a pseudofunctor

$$T: \Delta \longrightarrow K,$$

a descent object is an object $\text{Desc } T$ of K equipped with a Δ -natural isomorphism

$$K(K, \text{Desc } T) \cong \text{Desc } K(K, T).$$

A codescent object in K for a pseudofunctor

$$S: \Delta^{\text{op}} \longrightarrow K \text{ is a descent object for}$$

$$S^{\text{op}}: \Delta \longrightarrow K^{\text{op}}.$$

Proposition 4.5 A descent object for a functor $T: \Delta \rightarrow K$ is a limit for T weighted by the inclusion $\Delta \rightarrow \text{Cat}$.

Bénabou - Roubaud [Comptes Rendue 1970] and
 for Beck [unpublished] expressed descent data
 in terms of Eilenberg-Moore algebras for monads.
 We shall describe a version of this.

A functor $P: \mathcal{X} \rightarrow \mathcal{C}$ is called an
opfibration (originally Grothendieck used the
 term "cofibration" and some authors persist
 with this) when $P^{\text{op}}: \mathcal{X}^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ is a
 fibration. Cartesian morphisms and inverse
 images for P^{op} are called opcartesian
morphisms and direct images for P .

Proposition 4.6 If a cloven fibration $P: \mathcal{X} \rightarrow \mathcal{C}$
is also an opfibration then the inverse
image functor

$$h^*: \mathcal{X}^u \rightarrow \mathcal{X}^v$$

for each $h: v \rightarrow u$ in $\mathcal{C}$, has a left
adjoint defined by direct image $\underset{\wedge}{h_*}$ along h .

Proof $\mathcal{X}^v(Y, h^*X) \cong \{x: Y \rightarrow X \mid Px = h\}$
 $\cong \mathcal{X}^u(h_*Y, X). \quad \square$

For any functor $P: \mathcal{X} \rightarrow \mathcal{C}$, consider a commutative square

$$\begin{array}{ccc} R & \xrightarrow{q} & Y \\ p \downarrow & & \downarrow y \\ X & \xrightarrow{x} & Z \end{array} \quad (4.5)$$

in $\mathcal{X}$. The following condition is attributed to Chevalley in [Bénabou-Roubaud] and to Beck in other places.

Chevalley-Beck condition If (4.5) is a pullback in $\mathcal{X}$, preserved by P , with x cartesian and y opcartesian, then p is opcartesian.

By Lemma 2.1, the hypothesis here that (4.5) be a pullback in $\mathcal{X}$ is equivalent to the assumption that q be cartesian.

Example 4.7 The Chevalley-Beck condition holds for $\text{cod}: \mathcal{C}^{\mathcal{C}} \rightarrow \mathcal{C}$ for any category $\mathcal{C}$. To see this, notice that a square (4.5) in $\mathcal{C}^{\mathcal{C}}$ is a cube

$$\begin{array}{ccccc} R & & & & Y \\ & \searrow & & \nearrow & \\ & M & \xrightarrow{q_1} & V & \\ p_1 \downarrow & & & & \downarrow y_1 \\ & U & \xrightarrow{x_1} & W & \\ & \nearrow & & \searrow & \\ X & & & & Z \end{array} \quad \begin{array}{c} \xrightarrow{q_0} \\ \downarrow p_0 \\ \xrightarrow{x_0} \end{array} \quad (4.6)$$

The assumptions of Chevalley-Beck are that front, back, top and bottom faces of (4.6) are pullbacks, and that y_0 is invertible. Since the top face is a pullback, p_0 is also invertible. So p is opcartesian.

If P is a cloven fibration and opfibration, the Chevalley-Beck condition is the requirement that, for all pullback squares

$$\begin{array}{ccc} M & \xrightarrow{q} & V \\ p \downarrow & & \downarrow k \\ U & \xrightarrow{h} & W \end{array} \quad (4.7)$$

in $\mathcal{C}$, the canonical natural transformation

$$\begin{array}{ccc} \mathcal{X}^M & \xrightarrow{q_*} & \mathcal{X}^V \\ p^* \uparrow & \searrow \gamma & \uparrow k^* \\ \mathcal{X}^U & \xrightarrow{h_*} & \mathcal{X}^W \end{array} \quad (4.8)$$

is invertible. Under these circumstances, each category C in $\mathcal{C}$ determines a monad T_C on the category $\mathcal{X}^{C_0}$ as follows.

(35)

The endofunctor for the monad is the composite

$$\mathcal{X}^{C_0} \xrightarrow{d_0^*} \mathcal{X}^{C_1} \xrightarrow{d_1^*} \mathcal{X}^{C_0}$$

The unit is the composite natural transformation

$$1 \cong (d_1 i)_* (d_0 i)^* \cong d_{1*} i_* i^* d_0^* \xrightarrow{d_{1*} \varepsilon d_0^*} d_{1*} d_0^*$$

where $\varepsilon : i_* i^* \rightarrow 1$ is the counit for $i_* \dashv i^*$.

The multiplication is the composite natural transformation

$$\begin{aligned} d_{1*} d_0^* d_{1*} d_0^* &\cong_{d_{1*} \delta d_0^*} d_{1*} d_{2*} d_0^* d_0^* \cong (d_1 d_2)_* (d_0 d_0)^* \\ &\cong (d_1 d_1)_* (d_0 d_1)^* \cong d_{1*} d_{1*} d_1^* d_0^* \xrightarrow{d_{1*} \varepsilon d_0^*} d_{1*} d_0^* \end{aligned}$$

using the δ^{-1} of (4.8) for the pullback of section 1 (C4) and the counit ε for $d_{1*} \dashv d_1^*$.

Proposition 4.8 [SPub13; Prop. (9.10)] If $P: \mathcal{X} \rightarrow \mathcal{C}$ is a cloven fibration and opfibration satisfying the Chevalley - Beck condition then, for each category C in $\mathcal{C}$ there is an equivalence of categories

$$\mathcal{X}^C \underset{\text{-hand}}{\cong} (\mathcal{X}^{C_0})^{T_C}$$

where the right side is the category of Eilenberg - Moore algebras for the monad T_C .

(36)

Proof Just as for descent data with (4.1), for each $X \in \mathcal{X}^C$, we obtain a unique

$$e: d_0^* X_0 \longrightarrow d_1^* X_0 \quad (4.9)$$

in $\mathcal{X}^{C_1}$ defined by

$$\begin{array}{ccc} & e \nearrow d_1^* X_0 & \searrow d_1^{X_0} \\ d_0^* X_0 = X_1 & \xrightarrow{d_1} & X_0 \end{array}$$

Diagrams as in (4.2) and (4.3) hold yielding that the mate

$$\hat{e}: T_c X_0 = d_{1*} d_0^* X_0 \longrightarrow X_0,$$

of e under the adjunction $d_{1*} \dashv d_1^*$, is a T_c -algebra structure on X_0 . We leave the remainder to the reader. $\square$

Example 4.9 For the moment, consider $\mathbf{Cat}$ as a category and let us look at the fibration

$$\text{cod}: \mathbf{Cat}^2 \longrightarrow \mathbf{Cat}.$$

The cartesian morphisms for a cleavage are the chosen pullbacks in $\mathbf{Cat}$. The chosen opcartesian morphisms over $H: \mathcal{C} \rightarrow \mathcal{D}$ are commutative squares of the form

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{1_{\mathcal{X}}} & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathcal{C} & \xrightarrow{H} & \mathcal{D} \end{array}$$

The Chevalley-Beck condition holds as we saw in Example 4.7. For any category $\mathcal{C}$, we have a category

$$\text{sq}\mathcal{C} : \mathcal{C}^{\mathcal{B}} \xrightleftharpoons[\text{cod}]{\text{comp}} \mathcal{C}^{\mathcal{D}} \xrightleftharpoons[\text{cod}]{\text{id}} \mathcal{C}$$

in Cat obtained as

$$\Delta^{\text{op}} \xrightarrow{\text{incl}} \text{Cat}^{\text{op}} \xrightarrow{[-, \mathcal{C}]} \text{Cat}.$$

The reason for the name $\text{sq}\mathcal{C}$ is because it is the "double category of squares" in the category $\mathcal{C}$. Proposition 4.8 yields an equivalence of categories

$$(\text{Cat}^{\mathcal{D}})^{\text{sq}\mathcal{C}} \simeq (\text{Cat}/\mathcal{C})^{T_{\text{sq}\mathcal{C}}}. \quad (4.10)$$

The monad $T_{\text{sq}\mathcal{C}}$ is easily described. The endofunctor of $\text{Cat}/\mathcal{C}$ takes a functor $P : \mathcal{X} \rightarrow \mathcal{C}$ to the split opfibration

$$\text{cod} : P/\mathcal{C} \longrightarrow \mathcal{C}$$

with domain the comma category $P/\mathcal{C}$

(38)

of the functors $P: \mathcal{X} \rightarrow \mathcal{C}$ and $1_{\mathcal{C}}: \mathcal{C} \rightarrow \mathcal{C}$.
 The objects of $P/\mathcal{C}$ are (X, h, U) where $X \in \mathcal{X}$
 and $h: PX \rightarrow U$ in $\mathcal{C}$. The multiplication of the
 monad $\text{at. } P$
 $\wedge$ is the functor

$$\begin{array}{ccc} \text{cod}/\mathcal{C} & \xrightarrow{\mu} & P/\mathcal{C} \\ & \searrow \text{cod} & \swarrow \text{cod} \\ & \mathcal{C} & \end{array}$$

taking (X, h, U, r, V) to (X, rh, V) and
 the unit of the monad $\text{at. } P$
 $\wedge$ is the functor

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{\eta} & P/\mathcal{C} \\ & \searrow P & \swarrow \text{cod} \\ & \mathcal{C} & \end{array}$$

taking X to $(X, 1_{PX}, PX)$. In fact, the
 categories in (4.10) are 2-categories in an
 obvious way ($T_{sq}\mathcal{C}$ is a 2-monad on $\text{Cat}/\mathcal{C}$)
 and the equivalence is an equivalence of
 2-categories. The $T_{sq}\mathcal{C}$ -algebras are easily
 identified as split opfibrations $P: \mathcal{X} \rightarrow \mathcal{C}$;
 the $T_{sq}\mathcal{C}$ -action is defined by the functor

$$P/\mathcal{C} \xrightarrow{\alpha} \mathcal{X}$$

taking (X, h, U) to $h_*X \in \mathcal{X}^U$. If we

write $\text{Spl}_{\text{op}} \mathcal{C}$ for the 2-category of split
 offibrations over $\mathcal{C}$, cleavage preserving
 functors over $\mathcal{C}$, and natural transformations
 over $\mathcal{C}$, we see that (4.10) can be extended
 to equivalences of 2-categories

$$(\text{Cat}^2)^{\text{sp} \mathcal{C}} \simeq (\text{Cat}/\mathcal{C})^{\text{Type}} \simeq \text{Spl}_{\text{op}} \mathcal{C} \simeq [\mathcal{C}, \text{Cat}]. \quad (4.11)$$

Example 4.10 For any category B in $\mathcal{C}$, the
 codescent object of the simplicial object

$$B_2 \begin{array}{c} \xrightarrow{d_0} \\ \xrightarrow{d_1} \\ \xrightarrow{d_2} \end{array} B_1 \begin{array}{c} \xrightarrow{d_0} \\ \xleftarrow{i} \\ \xrightarrow{d_1} \end{array} B_0 \quad (4.12)$$

in $\text{Cat } \mathcal{C}$ is easily seen to be B itself.

For, the definition is at the beginning of section 1
 make it clear that

$$(\text{Cat } \mathcal{C})(B, C)$$

is the descent category for the simplicial category

$$(\text{Cat } \mathcal{C})(B_0, C) \rightleftarrows (\text{Cat } \mathcal{C})(B_1, C) \rightrightarrows \dots$$

Observe too that, for each U in $\mathcal{C}$, the nerve
 of the category $\mathcal{C}(U, B)$ is

$$\dots \rightrightarrows \mathcal{C}(U, B_2) \rightrightarrows \mathcal{C}(U, B_1) \begin{array}{c} \xrightarrow{\mathcal{C}(U, d_0)} \\ \xleftarrow{\mathcal{C}(U, d_1)} \end{array} \mathcal{C}(U, B_0);$$

so the codescent object of (4.12) is preserved (40)
 by the 2-functors $\mathcal{C}(U, -) : \text{Cat } \mathcal{C} \rightarrow \text{Cat}$.
 This provides another proof of Corollary 2.4
 that $\mathcal{C} \rightarrow \text{Cat } \mathcal{C}$ is dense by showing that
 there is a density presentation via a codescent construction.

§5. Split fibrations in a category

Let $\mathcal{C}$ be a category with pullbacks.
 For each category B in $\mathcal{C}$, we can construct
 a category B^2 in $\mathcal{C}$ as follows. Form the
 pullback

$$\begin{array}{ccccc}
 & & P & & \\
 & p \swarrow & & \searrow q & \\
 B_2 & & M & & B_2 \\
 d_0 \swarrow & & M & & \swarrow d_1 \\
 B_1 & & B_1 & & B_1 \\
 & d_1 \searrow & & \swarrow d_2 &
 \end{array}$$

yielding the graph

$$P \begin{array}{c} \xrightarrow{d_0 p} \\ \xrightarrow{d_2 q} \end{array} B_1$$

which underlies our category B^2 . There is
 a graph isomorphism

$$\mathcal{C}(U, B^2) \cong \mathcal{C}(U, B)^2 \quad (5.1)$$

natural in U . The right-hand side of (5.1) has
 a category structure: the arrow category of

$\mathcal{C}(U, B)$. Since pullbacks exist in $\mathcal{C}$, we can ^(4.1) construct the objects required to complete B^2 to a simplicial object in $\mathcal{C}$, and by Yoneda we transport the morphisms across (5.1) and internalize to $\mathcal{C}$. Using Example 4.9, we extend the isomorphism (5.1) to a isomorphism

$$(\text{Cat } \mathcal{C})(A, B^2) \cong (\text{Cat } \mathcal{C})(A, B)^2, \quad (5.2)$$

2-natural in A . This says that B^2 is the cotensor of 2 and B in $\text{Cat } \mathcal{C}$.

The 2-category $\text{Cat } \mathcal{C}$ has pullbacks formed pointwise with the simplicial objects. We can therefore compose the span

$$B \xleftarrow{d_0} B^2 \xrightarrow{d_1} B$$

with itself; for example, we obtain

$$\begin{array}{ccccc} & & B^{\mathbb{B}} & & \\ & \swarrow d_0 & & \searrow d_2 & \\ & B^2 & \text{pb} & B^2 & \\ \swarrow d_0 & & & & \searrow d_1 \\ B & & B & \xleftarrow{d_0} & B \end{array}$$

and then a functor $d_1: B^{\mathbb{B}} \rightarrow B^2$ lifting $d_1: B_2 \rightarrow B_1$. This gives a category

$$\text{sq } B: \quad B^{\mathbb{B}} \begin{array}{c} \xrightarrow{d_0} \\ \xrightarrow{d_1} \\ \xrightarrow{d_2} \end{array} B^2 \begin{array}{c} \xrightarrow{d_0} \\ \xleftarrow{d_1} \end{array} B$$

in $\text{Cat } \mathcal{C}$ such that there is a natural

isomorphism

$$\mathcal{C}(U, sq B) \cong sq \mathcal{C}(U, B)$$

where the right-hand side is explained in Example 4.9. Using Example 4.10, we extend the isomorphism to

$$(\text{Cat } \mathcal{C})(A, sq B) \cong sq(\text{Cat } \mathcal{C})(A, B). \quad (5.3)$$

Looking at the fibration

$$\text{cod} : (\text{Cat } \mathcal{C})^2 \longrightarrow \text{Cat } \mathcal{C}$$

and the category $sq B$ in $\text{Cat } \mathcal{C}$, we obtain a category

$$\text{Spl}_{sq B} := ((\text{Cat } \mathcal{C})^2)^{sq B}.$$

By Proposition 4.8, we have an equivalence

$$\text{Spl}_{sq B} \simeq ((\text{Cat } \mathcal{C})/B)^{T_{sq B}}. \quad (5.4)$$

The monad $T_{sq B}$ is easily identified. For any functor $f: X \rightarrow B$ in $\mathcal{C}$, form the pullback

$$\begin{array}{ccccc} f/B & \xrightarrow{\quad} & B^2 & \xrightarrow{d_1} & B \\ d_0 \downarrow & \text{pb} & \downarrow d_0 & & \\ X & \xrightarrow{f} & B & & \end{array}$$

The endofunctor of $T_{sq B}$ takes $f: X \rightarrow B$ to $d_1: f/B \rightarrow B$. We see that $T_{sq B}$ is actually a 2-monad on $(\text{Cat } \mathcal{C})/B$ yielding a

2-category $\text{Spl}_{\text{op}} B$.

The objects of $\text{Spl}_{\text{op}} B$ are called split opfibrations over B in $\mathcal{C}$. Such an object is a functor $p: E \rightarrow B$ in $\mathcal{C}$ equipped with a functor $d_1: p/B \rightarrow E$ for which there exists a category structure in $(\text{Cat } \mathcal{C})^2$ extending the diagram

$$\begin{array}{ccc}
 p/B & \begin{array}{c} \xrightarrow{d_0} \\ \xleftarrow{i} \\ \xrightarrow{d_1} \end{array} & E \\
 \downarrow & & \downarrow p \\
 B^2 & \begin{array}{c} \xrightarrow{d_0} \\ \xleftarrow{i} \\ \xrightarrow{d_1} \end{array} & B
 \end{array} \quad (5.5)$$

The functor i in the top line of (5.5) is defined by the requirement that the pasted composite 2-cell

$$\begin{array}{ccccc}
 & & E & & \\
 & \swarrow & \downarrow i & \searrow & \\
 1 & & p/B & & p \\
 & \swarrow d_0 & \xrightarrow{\lambda} & \searrow d_1 & \\
 & E & \xrightarrow{p} & B &
 \end{array} \quad (5.6)$$

should be the identity 2-cell of p . In fact, the action $d_1: p/B \rightarrow E$ of the monad $T_{\text{op}} B$ is unique up to isomorphism since one can show that is left adjoint $d_1 \dashv i$ to i .

We already defined discrete offibrations in section 3. Replacing C by B in the pullback (3.1), we can deduce the pullback

$$\begin{array}{ccc} E^2 & \xrightarrow{d_0} & E \\ p^2 \downarrow & & \downarrow p \\ B^2 & \xrightarrow{d_0} & B \end{array} \quad (5.7)$$

in $\text{Cat } \mathcal{C}$. Define $j: E^2 \rightarrow p/B$ by the equality

$$\begin{array}{c} \begin{array}{ccccc} & E^2 & & & \\ & \downarrow j & \searrow d_1 & & \\ & p/B & & E & \\ & \swarrow d_0 & \xrightarrow{\lambda} & \searrow d_1 & \downarrow p \\ E & \xrightarrow{p} & B & & \end{array} \\ \downarrow d_0 \quad \quad \quad \downarrow p \end{array} = \begin{array}{c} E^2 \\ \downarrow d_0 \left(\begin{array}{c} \xrightarrow{\lambda} \\ \xrightarrow{\lambda} \end{array} \right) \downarrow d_1 \\ E \\ \downarrow p \\ B \end{array} \quad (5.8)$$

It follows that a functor $p: E \rightarrow B$ in $\mathcal{C}$ is a discrete offibration if and only if j is invertible. We can see then that p is indeed an offibration by taking the d_1 of (5.5) to be the composite

$$p/B \xrightarrow{j^{-1}} E^2 \xrightarrow{d_1} E. \quad (5.9)$$

We can also then see that every commutative

triangle

(45)

$$\begin{array}{ccc} E & \xrightarrow{f} & F \\ & \searrow p & \swarrow q \\ & B & \end{array} \quad (5.10)$$

in $\text{Cat } \mathcal{C}$, with q a discrete offibration and p any split offibration, is a morphism in $\text{Sp}_{\text{op}} B$. Write $\text{Disc}_{\text{op}} B$ for the full subcategory of $\text{Sp}_{\text{op}} B$ whose objects are the discrete offibrations. Our observations show that the 2-functor

$$\text{Disc}_{\text{op}} B \xrightarrow{\text{incl.}} \text{Sp}_{\text{op}} B \xrightarrow{\text{und.}} (\text{Cat } \mathcal{C})/B \quad (5.11)$$

is fully faithful.

in $\mathcal{C}$.

Let $p: E \rightarrow C$ be a discrete offibration.

Recall the definition of $\bar{p}^u: \mathcal{C}(u, C) \rightarrow \mathcal{C}/u$ in Section 3 (3.4). For each category B in $\mathcal{C}$, we have a morphism of diagrams

$$\begin{array}{ccccc} \mathcal{C}(B_0, C) & \xrightleftharpoons[\mathcal{C}(d_1, 1)]{\mathcal{C}(d_0, 1)} & \mathcal{C}(B_1, C) & \xrightleftharpoons[\mathcal{C}(d_2, 1)]{\mathcal{C}(d_0, 1)} & \mathcal{C}(B_2, C) \\ \bar{p}^{B_0} \downarrow & & \downarrow \bar{p}^{B_1} & & \downarrow \bar{p}^{B_2} \\ \mathcal{C}/B_0 & \xrightleftharpoons[d_1^*]{d_0^*} & \mathcal{C}/B_1 & \xrightleftharpoons[d_2^*]{d_0^*} & \mathcal{C}/B_2 \end{array} \quad (5.12)$$

in the "pseudo" sense. This induces a functor

$$\bar{p}^B : (\text{Cat } \mathcal{C})(B, C) \longrightarrow \text{Spl}_{pB} \quad (5.13)$$

on the descent categories. Clearly then:

Proposition 5.1 If $p: E \rightarrow C$ is an internal full subcategory of $\mathcal{C}$ then, for all categories B in $\mathcal{C}$, the functors $\bar{p}^B$ of (5.13) are fully faithful.

Corollary 5.2 If $\mathcal{C}$ is a finitely complete, cartesian closed category and $p: E \rightarrow C$ is an internal full subcategory then the 2-category $\text{Cat } \mathcal{C}$ equipped with the usual duality involution interchanging d_0 and d_1 , and the classifying discrete opfibration p , is a 2-topos in the sense of [Weber APCS 2007].

In Section 5 of [Weber APCS 2007] it is shown how to construct a Yoneda structure [St-Walt JAlg 1978] which is "good" in Weber's sense.

§6. Internal full subcategories of $\mathcal{C}at$

An algorithm for finding internal full subcategories of a locally presentable category $\mathcal{C}$ was provided in [St. Pub. 13]. This was applied to $\mathcal{C} = \mathcal{C}at$ in Section 8.10 of that paper. However, we shall give an internal full subcategory without going through the discovery process.

For a (small) category A , write $\hat{A}$ for the category $[A^{op}, \mathbf{Set}]$ of contravariant set-valued functors on A . It is the small colimit completion of A in the sense that restriction along the Yoneda embedding $y_A: A \rightarrow \hat{A}$ yields an equivalence of categories

$$\mathbf{Cocts}(\hat{A}, X) \simeq [A, X], \quad (6.1)$$

where the left side is the full subcategory of $[\hat{A}, X]$ consisting of the small-colimit-preserving functors into the small cocomplete category X .

Let $\mathbf{mod}_0$ denote the category whose objects are small categories A and whose morphisms $m: A \rightarrow B$ are colimit-preserving

functors $m : \hat{A} \rightarrow \hat{B}$. Each morphism $m : A \rightarrow B$ determines a functor

$$\bar{m} : B^{\text{op}} \times A \longrightarrow \text{Set}$$

defined by

$$\bar{m}(b, a) = m(A(-, a))(b). \quad (6.2)$$

By (6.1), m is determined uniquely up to isomorphism by $\bar{m}$. For each $\varphi \in \hat{A}$, we have a natural family

$$\kappa : \varphi(a) \times \bar{m}(b, a) \longrightarrow m(\varphi)(b)$$

defined by

$$\kappa(x, y) = m(\hat{x})_b(y) \quad (6.3)$$

where $\hat{x} : A(-, a) \rightarrow \varphi$ is unique with

$$\hat{x}_a(1_a) = x \in \varphi(a).$$

In particular, for $m : A \rightarrow B$ and $n : B \rightarrow C$ in mod , we have

$$\kappa : \bar{m}(b, a) \times \bar{n}(c, b) \longrightarrow \overline{nm}(c, a),$$

which is a universal extraordinary natural family in the variable b ; in other words,

$$\int^b \bar{m}(b, a) \times \bar{n}(c, b) \cong \overline{nm}(c, a). \quad (6.4)$$

(49)

Let $T: X \rightarrow \text{mod}_0$ be a functor and define a category $El(T)$ as follows. The objects are pairs (x, a) where $x \in X$ and $a \in Tx$. Morphisms $(\xi, \tau): (x, a) \rightarrow (y, b)$ consist of $\xi: x \rightarrow y$ in X and $\tau \in \overline{T\xi}(b, a)$. Composition is defined by

$$\begin{array}{ccc} & (\xi, \tau) \rightarrow (y, b) & (\eta, v) \\ (x, a) & \searrow & \searrow \\ & (z, c) & \end{array} \quad (6.5)$$

$$\xrightarrow{(\eta\xi, v*\tau)}$$

where $v*\tau$ is the value of the function

$$\kappa: \overline{T\xi}(b, a) \times \overline{T\eta}(c, b) \rightarrow \overline{T(\eta\xi)}(c, a)$$

at (τ, v) . The identity morphism of (x, a) is $(1_x, 1_a)$ where we note that

$$\overline{T1_x}(a, a) = (Tx)(a, a).$$

There is a projection functor

$$p_T: El(T) \rightarrow X$$

defined by $p_T(x, a) = x$ and $p_T(\xi, \tau) = \xi$.

Let mod_1 denote the category whose objects are functors $f: A \rightarrow B$ between small categories. A morphism is a square

$$\begin{array}{ccc}
 A & \xrightarrow{f} & B \\
 m \downarrow & \Rightarrow \theta & \downarrow n \\
 C & \xrightarrow{g} & D
 \end{array} \quad (6.6)$$

where f and g are the domain and codomain functors, where m and n are morphisms of mod_0 , and where θ is a natural family of functions

$$\theta_{b,a} : \bar{m}(c, a) \longrightarrow \bar{n}(gc, fa).$$

There is a bijection between such θ and natural transformations θ as in the diagram

$$\begin{array}{ccc}
 \hat{A} & \xleftarrow{\hat{f}} & \hat{B} \\
 m \downarrow & \Rightarrow \theta & \downarrow n \\
 \hat{C} & \xleftarrow{\hat{g}} & \hat{D}
 \end{array} \quad (6.7)$$

Composition is achieved by vertical pasting of squares of the form (6.7).

We have functors

$$d_0, d_1 : \text{mod}_1 \longrightarrow \text{mod}_0 \quad (6.8)$$

defined by $d_0(f) = A$, $d_0(m, \theta, n) = m$,
 $d_1(f) = B$, $d_1(m, \theta, n) = n$ (referring to (6.6)).

(51)

This defines a graph in Cat . There is also a canonical structure of category in Cat having (6.8) as underlying graph. Composition for this category mod in Cat is derived from horizontal pasting of squares of the form (6.7).

We are at present regarding Cat as a category, not a 2-category. When we write $\text{Cat}(X, \text{mod})$ we mean in the sense of $\mathcal{C}(U, C)$ as described at the beginning of section 1.

We shall now extend the construction summarized by (6.5) to a functor

$$E_1 : \text{Cat}(X, \text{mod}) \rightarrow \text{Cat}/X. \quad (6.9)$$

Let T and S be objects of $\text{Cat}(X, \text{mod})$; that is, T and $S : X \rightarrow \text{mod}_0$ are functors. A morphism $\Theta : T \rightarrow S$ in $\text{Cat}(X, \text{mod})$ is a functor $\Theta : X \rightarrow \text{mod}$, with $d_0 \Theta = T$ and $d_1 \Theta = S$. Each morphism $\Xi : x \rightarrow y$ in X yields a diagram

$$\begin{array}{ccc} Tx & \xrightarrow{\Theta_x} & Sx \\ T\Xi \downarrow & \Theta_\Xi \Rightarrow & \downarrow S\Xi \\ Ty & \xrightarrow{\Theta_y} & Sy \end{array}$$

in which $\oplus_x$ and $\oplus_y$ are functors. Define a functor (52)

$$\begin{array}{ccc} E(T) & \xrightarrow{E(\oplus) = F} & E(S) \\ & \searrow p_T & \swarrow p_S \\ & X & \end{array} \quad (6.10)$$

over X as follows:

$$F(x, a) = (x, \oplus_x(a)), \quad F(\xi, \tau) = (\xi, \theta_\xi(\tau))$$

using $\theta_\xi : \overline{T\xi}(b, a) \rightarrow \overline{S\xi}(\oplus_y(b), \oplus_x(a))$.

By looking at (6.5) and using the fact that $\oplus$ is a functor, we see that F is a functor. Clearly (6.10) commutes. Composition in $\text{Cat}(X, \text{mod})$ involves horizontal pasting of squares (6.7), from which we see that we do have a functor (6.9).

The following is related to Gray's Yoneda-like lemma [Gray, SLNM 99(1969); see page 290]; also see [Kelly, SLNM 420(1974); see page 210].

Theorem 6.1 The functors EL of (6.9) are
fully faithful for all X . The family is
pseudonatural in X .

Proof The proof of the first sentence is a mere re-tracing of the steps in the definition of EL on morphisms. The second sentence follows from the observation that, for each functor $r: Y \rightarrow X$, we have a pullback square

$$\begin{array}{ccc} EL(Tr) & \longrightarrow & EL(T) \\ p_{Tr} \downarrow & & \downarrow p_T \\ Y & \xrightarrow{r} & X. \quad \square \end{array}$$

Corollary 6.2 The family of functors (6.9) exhibits mod as an internal full subcategory of Cat .

By the generalized Yoneda Lemma of Section 2, the pseudonatural family (6.9) is determined by a discrete opfibration

$$p: \text{obj} \longrightarrow \text{mod} \quad (6.11)$$

between categories in Cat . We shall describe obj .

The category obj_0 has objects pairs (A, a) where A is a small

category and a is an object of A . A morphism $(m, \mu) : (A, a) \rightarrow (B, b)$ in obj_0 consists of a morphism $m : A \rightarrow B$ in mod_0 together with $\mu \in \bar{m}(b, a)$. Composition is a special case of (6.5) and uses the κ of (6.4). We have the projection functor

$$p_0 : \text{obj}_0 \rightarrow \text{mod}_0 \quad (6.12)$$

taking $(m, \mu) : (A, a) \rightarrow (B, b)$ to $m : A \rightarrow B$.

Proposition 6.3 The functor p_0 of (6.12) is
powerful in the category Cat . That is, the
functor

$$p_0^* : \text{Cat}/\text{mod}_0 \rightarrow \text{Cat}/\text{obj}_0$$

has a right adjoint.

Proof We must show that the functor p_0 has the factorization lifting property of Giraud-Conduché [Giraud], [Conduché], [St-V].

Take a composable pair

$$A \xrightarrow{m} B \xrightarrow{n} C$$

in mod_0 and a lifting

$$(A, a) \xrightarrow{(nm, \lambda)} (C, c)$$

to obj_0 of the composite nm . By (6.4), there exists $b \in B$ and $(\mu, \nu) \in \bar{m}(b, a) \times \bar{n}(c, b)$ such that $\kappa(\mu, \nu) = \lambda$. This gives a factorization

$$(A, a) \xrightarrow{(m, \mu)} (B, b) \xrightarrow{(n, \nu)} (C, c)$$

of (nm, λ) which, using (6.4) again, determines a unique path component in the category of such liftings, as required. $\square$

Now we shall define the category obj_1 . The objects are pairs (f, a) where $f: A \rightarrow B$ is a functor and $a \in A$. A morphism

$$(m, \mu, n, \theta) : (f, a) \rightarrow (g, c) \quad (6.13)$$

consists of a morphism $(m, n, \theta) : f \rightarrow g$ as in (6.6) and $\mu \in \bar{m}(c, a)$. Composition is such that we have the functor

$$P_1 : \text{obj}_1 \rightarrow \text{mod}_1 \quad (6.14)$$

taking (6.13) to (m, n, θ) . We also have

the functor

$$d_0 : \text{obj}_1 \longrightarrow \text{obj}_0, \quad (6.15)$$

taking (6.13) to $(m, \mu) : (A, a) \longrightarrow (C, c)$,
and the functor

$$d_1 : \text{obj}_1 \longrightarrow \text{obj}_0 \quad (6.16)$$

taking (6.13) to $(n, \theta_{c,a}(\mu)) : (B, fa) \longrightarrow (D, gc)$.

By the general theory in Section 2, the functors d_0 and d_1 of (6.15) and (6.16) are the underlying graph in Cat of a category obj in Cat , and the functors p_0 and p_1 of (6.12) and (6.14) form a functor

$$p : \text{obj} \longrightarrow \text{mod}$$

in Cat . Indeed, we also know that p is a discrete offibration pullback along which gives the functor El of (6.9).

Categories in Cat are called double categories. They were defined by Ehresmann []. Another reference is [Kelly-StSLNM420]. We write Dbl for the 2-category

(57)

$\text{Cat}(\text{Cat})$ of double categories. Actually, looking only at this 2-category structure on Db loses quite a bit of the symmetry of the situation. Since Cat is cartesian closed, so too is Db and that is important from the viewpoint of the 2-topos structure. It also means that Db is hom-enriched in itself. There are two underlying functors

$$\text{Db} \longrightarrow \text{Cat}$$

which induce two 2-functors

$$\text{Db-Cat} \longrightarrow \text{Cat-Cat} = 2\text{-Cat}$$

whose values at Db itself give two ways to regard Db as a 2-category.